

Bhuvan Patil

Mumbai, Maharashtra , India | bhu1patil421@gmail.com | +91 9322239586 |

LinkedIn : bhuvan-patil | Github : BhuvanPatil

Summary

Recent graduate in Artificial Intelligence and Data Science, seeking roles in AI/ML Engineering, GenAI/LLM Engineering, or Data Science. Strong foundation in machine learning, deep learning, and Large Language Model/RAG systems through hands-on projects. No formal work experience yet, but genuinely curious about the field and quick to learn.

Education

Vivekanand Education Society's Institute of Technology B.E. in Artificial Intelligence and Data Science CGPA: 9.09	2022 – 2026
National Junior College Higher Secondary Examination Percentage: 84.5%	2020 – 2022
National English School Senior Secondary Examination Percentage: 90.80%	2011 – 2020

Skills

Languages: Python, Java, SQL

Machine Learning & Deep Learning: TensorFlow, PyTorch, Scikit-learn, Neural Networks, GANs, VAEs, Autoencoders (conceptual)

Generative AI & NLP: LLMs, LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Prompt Engineering, Hugging Face, Ollama

Development & Backend: FastAPI, REST APIs, Streamlit, React (basic)

Databases & Data: MySQL, PostgreSQL, Vector Databases (Qdrant, FAISS), Pandas, NumPy, Matplotlib

Tools & Deployment: Docker, Git, Supabase

Publications

VisioniX: Multimodal AI Assistant for Image Recognition and Conversational Reasoning • Published in VDI-Z Integrierte Produktion Journal, Vol. 13, Issue 4, April 2026 (Impact Factor: 5.3)	Link
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Experience

Agentic Multi-Document RAG System • Designed a 9-node LangGraph pipeline that routes each query to one of four strategies (document search, web search, LLM knowledge, or cache) using an LLM classifier with structured output, reaching 100% routing accuracy on a 20-question benchmark at 0.18s average latency • Built a hybrid retrieval layer combining dense (Qdrant MMR) and sparse (BM25) search merged via Reciprocal Rank Fusion, with parent-document expansion for full-context retrieval and cross-encoder reranking (BAAI/bge-reranker-large) to improve retrieval precision. • Implemented a self-correction loop: low-confidence retrievals trigger automatic query rewriting and retry, and insufficient context falls back to live web search, with retry limits to prevent infinite loops	Github
VisioniX (Team Project) • Team-built, full-stack AI platform that analyzes uploaded images using computer vision (YOLOv8, BLIP, CLIP, OCR) and lets users hold multi-turn conversations about them via a locally-hosted LLM, with support for training and auto-deploying custom vision models	Github

- Built the image-grounded conversational chat feature, enabling multi-turn Q&A where the LLM (Ollama, locally-hosted) answers based on the visual content of the uploaded image
- Built a browser-driven YOLOv8 fine-tuning pipeline that runs training jobs in the background and automatically evaluates model accuracy (mAP50/accuracy) against a configurable threshold, blocking deployment of models that don't meet the quality bar
- Automated the post-training deployment flow: models that pass the quality gate are uploaded to Hugging Face Hub and published as a live inference endpoint, instantly selectable as the active model in chat

Customer Churn Prediction

Github

- Built an end-to-end telecom churn prediction pipeline in Python using scikit-learn, with stratified train/test splitting to preserve class balance, MinMaxScaler for numeric features, and one-hot encoding for categorical features, wrapped in a single pipeline to prevent training/inference mismatch
- Built a FastAPI backend with Pydantic schema validation, enum-constrained inputs, and cross-field business-rule checks (e.g. rejecting invalid service combinations) before running predictions
- Built a Streamlit frontend that dynamically constrains form inputs based on prior selections and displays churn probability, predicted class, and confidence from the live API

Achievements & Extracurricular Activities

2nd Place, Hack-AI-thon — Built the LangGraph agent orchestration for a GEO analysis prototype (Team Project, 80+ teams)

Senior Graphics Officer, AI-CoLegion — Led graphics team; provided visual support for events and workshops.

Member, SORT Aesthetics — Designed props and visuals for social impact events.

1st Place, ILLUSION'25 — Served as Aesthetic Head of Team Intruders in an inter-collegiate cultural event.

Certifications

NVIDIA Deep Learning Institute

IBM Data Science Specialization

Ultimate MySQL Bootcamp